

Chapters 1-9 Warm-Up: Review

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1 Review Exercises

1. Write a program that produces the following table of contents:

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2. Write a program that collects information about a student:

```
score = int(input("Score: "))  
sub = input("Submitted? yes/no: ") == "yes"  
att = input("Attendance OK? yes/no: ") == "yes"
```

Create the Boolean variable

```
ready = sub and att
```

and use it in the following chained decision:

```
if score >= 90 and ready:  
    print("Excellent")  
elif score >= 70 and ready:  
    print("Pass")  
elif not sub or not att:  
    print("Incomplete")  
else:  
    print("Revise")
```

Run the program several times. Find inputs that cause each of the four branches to execute.

3. Define the following *diagonal* matrix:

```
A = [  
    [1, 0, 0],  
    [0, 1, 0],  
    [0, 0, 1]  
]
```

Suppose we did not know the matrix was diagonal. Create nested `for` loops and conditional logic that demonstrate the matrix is diagonal. For example:

```
is_diagonal = True  
for row in range(2):  
    for column in range(3):  
        if(...):  
            is_diagonal = False  
    ...
```

4. Define a function that evaluates $f(x) = \sin(x)^2 + \cos(x)^2$, and show that it is always 1, regardless of the input x .

5. Suppose a patient record is represented by a list:

```
patient = [1042, "Maria Garcia", 37]
```

The entries represent the patient ID, name, and age. Write a function

```
def display_patient(patient):  
    # your code here
```

that prints the record in the form

```
ID: 1042  
Name: Maria Garcia  
Age: 37
```

6. Several course records can be stored inside one master list:

```
courses = [  
    ["PHYS", 100, "SLC", 202],  
    ["MATH", 150, "SLC", 105],  
    ["BIOL", 101, "SLC", 210],  
    ["PHYS", 200, "SLC", 202]  
]
```

Lists can be sorted using `sort()`. Define

```
def sort_courses(courses):  
    courses.sort()
```

Then run

```
sort_courses(courses)  
for course in courses:  
    print(course)
```